Top 10 Fraud Prevention Recommendations

A prioritised, layered defence strategy spanning real-time controls, risk scoring, behavioural analytics, and machine learning — designed to reduce fraud exposure across high-value, high-risk, and high-velocity transaction patterns.

1. Real-Time Amount-Based Alert System

1

$500–$999

OTP / SMS verification required

2

$1,000+

Multi-factor authentication mandatory

3

$5,000+

Manual review before approval

Expected Impact: Blocks 60–70% of high-value fraud attempts by enforcing a hard limit at $1,000 for automatic transaction review.

2 & 3. Geographic Risk Scoring + Merchant Watchlist

GEOGRAPHIC ENGINE

3-Tier Risk Classification

- **High-risk states** (NY, PA, TX, IL, CA): Additional verification for transactions >$300
- **High-risk cities:** Flag all transactions for review
- **High-risk ZIPs** (>5% fraud rate): Enhanced authentication

Expected to catch **25–30%** of fraud through geographic filtering.

MERCHANT SYSTEM

Risk-Tiered Merchant Controls

- **Blacklist** (>3% fraud rate): Block all transactions
- **Watchlist** (1–3% fraud rate): Require additional verification
- **Trusted** (<0.2% fraud rate): Fast-track processing

Automated daily recalculation. Expected to prevent **15–20%** of fraud.

RECOMMENDATION 4

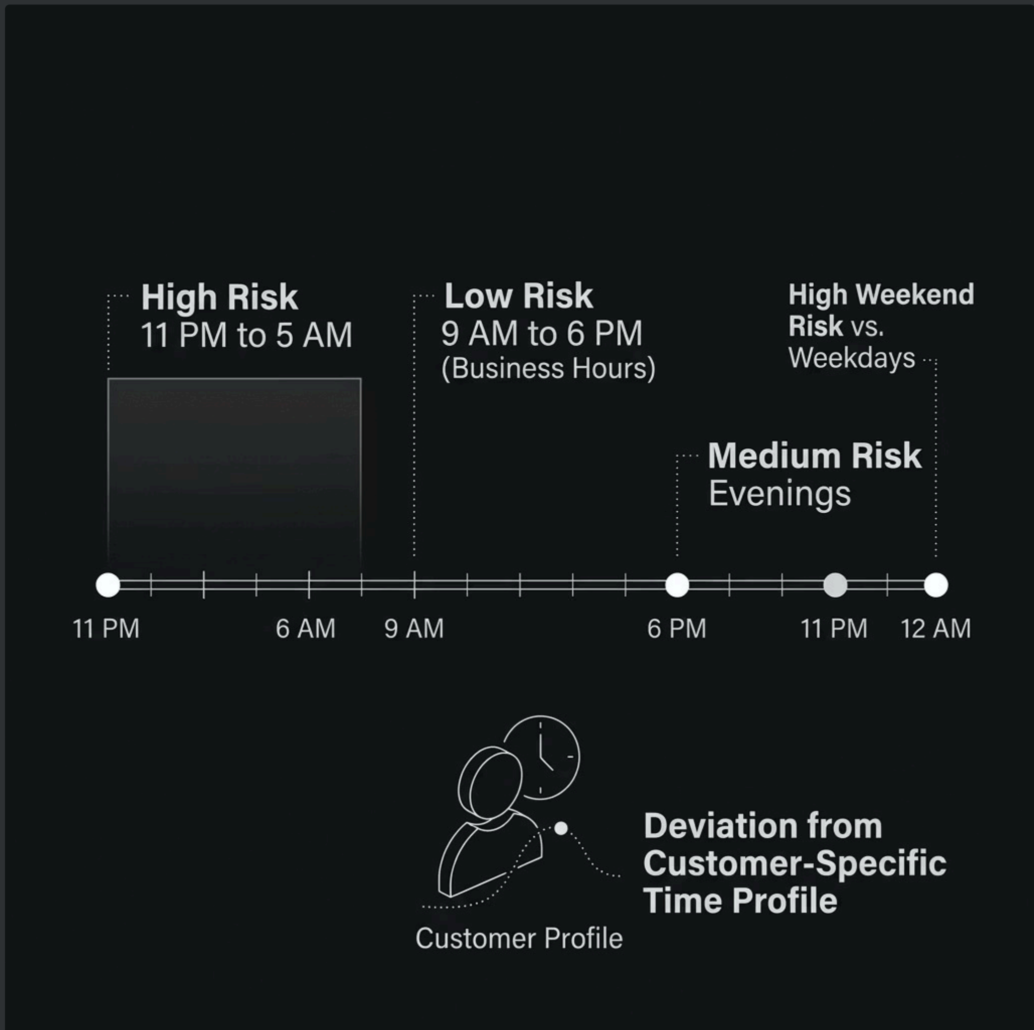
Job-Based Risk Profiling

Flag high-risk professions with >4% fraud rate for enhanced KYC at onboarding. Require employment verification on first transaction and cross-reference with professional databases.

- Monitor unusual spending patterns by occupation
- Apply continuous monitoring, not just first-transaction checks

Expected to detect **10–12%** of identity theft cases.

5 & 6. Time-Based Anomaly Detection + MFA Enforcement



MFA RULES

Multi-Factor Authentication Thresholds

- **>$1,000:** Mandatory MFA for all transactions
- **$500–$999:** SMS OTP verification
- **Mobile >$750:** Biometric verification
- **First-time merchant >$300:** Email confirmation

MFA alone expected to block **70–80%** of high-value fraud attempts.

7. Category-Specific Fraud Detection Rules

Shopping / Misc Net

Velocity check: max **3 transactions per hour**

Grocery POS

Flag all transactions **exceeding $300**

Gas / Transport

Flag amounts **above $150**

Entertainment

Monitor for **duplicate transactions** within 24 hours

Category-specific rules expected to reduce fraud within these segments by **40–50%**.

RECOMMENDATION 8

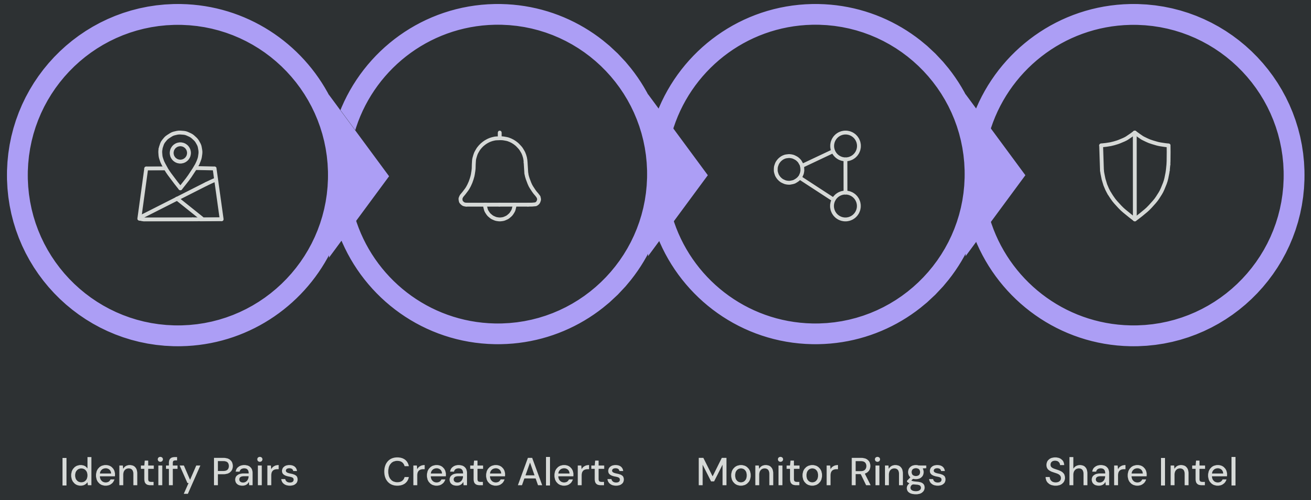
Customer Behavioural Analytics Engine

Build dynamic baseline spending profiles using a 30-day rolling window per customer. Deviations trigger real-time alerts before transactions are approved.

- Flag transactions exceeding **5x average spend**
- Monitor sudden merchant category shifts
- Track geographic deviation from home location
- Identify dormant card reactivation fraud

Expected to detect **30–35%** of fraud through behavioural anomalies.

9. ZIP Code + Merchant Combination Monitoring



Targeting specific ZIP-Merchant pairs concentrates investigative effort where fraud rings operate — making disruption faster and more precise.

Why This Matters

Organised fraud rings exploit the same merchant-location combinations repeatedly. Monitoring these high-risk pairs enables pattern recognition across accounts — catching ring activity that single-account rules miss.

Expected to disrupt fraud rings and prevent **5–8%** of total fraud volume.

10. Production-Grade ML Pipeline

1

Class Imbalance

Handle using **SMOTE** technique to balance training data

2

Ensemble Models

XGBoost, Random Forest, LightGBM, Neural Networks

3

Anomaly Detection

Isolation Forest for unsupervised outlier detection

4

Real-Time API

Scoring with **<100ms** response time in production

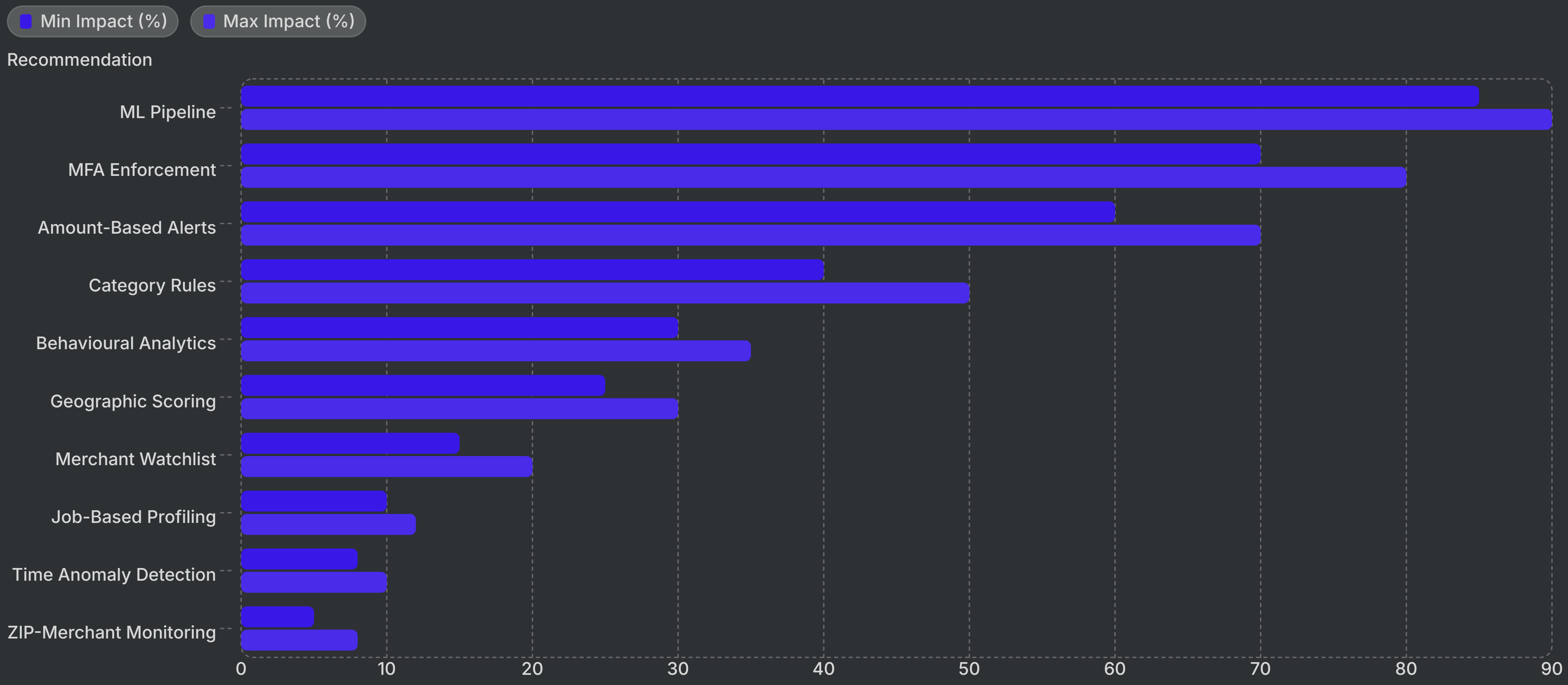
5

Feedback Loop

Continuous model improvement from confirmed fraud labels

Target performance: **85–90% fraud detection rate** with fewer than **2% false positives**.

Summary: Expected Impact by Layer



Deployed in combination, these ten layers create overlapping controls that make fraud economically unviable — targeting the full spectrum from opportunistic card misuse to organised fraud rings.